

Calculator Project

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Calculator

A dual-mode web calculator — basic and scientific — built with vanilla HTML, CSS, and JavaScript. Features glass-morphism design, animated particles, Orbitron typography, and full keyboard support.

HTML · CSS · JS



Design System



Typography

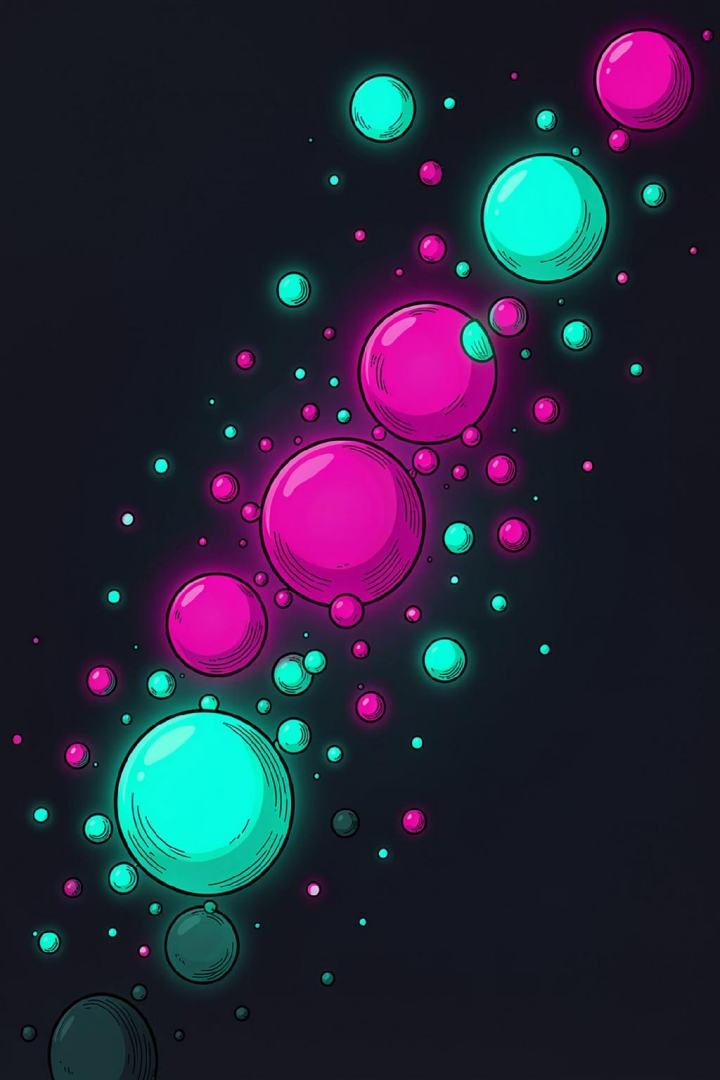
Orbitron for headings (900/700 weight),
Share Tech Mono for UI text – both
Google Fonts.

Color Palette

Cyan (#00c9ff), magenta (#ff4ecd), teal
(#00ff88) gradient text on dark navy
backgrounds.

Card Style

Glass-morphism: `backdrop-filter:`
`blur(20px)`, translucent background,
inner glow borders.



Background & Particles

Menu Background

Three overlapping radial gradients – cyan, magenta, teal – on `#0d0d1a` base.

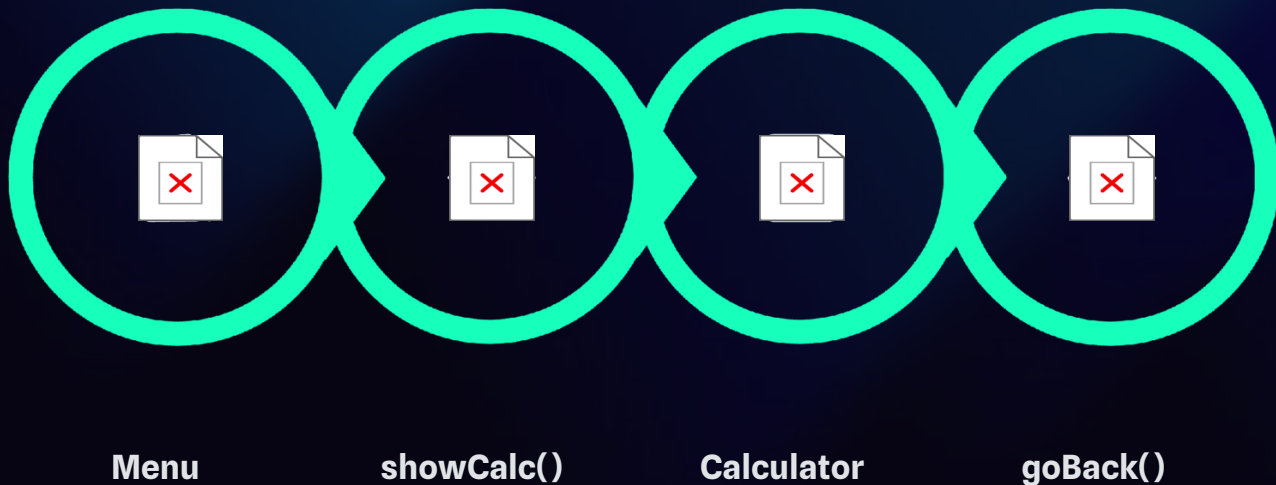
Calc Background

Darker `#040f1a` with subtle cyan/blue radial glows at corners.

Particle System

22 floating orbs, random sizes/colors, `floatUp` animation (8-22s loop), z-index 0.

Scene Transitions



The app toggles between three views by swapping `display` and swapping `body.className` between `menu-bg` and `calc-bg`. Card entry uses a `slideIn` keyframe animation (0.4s cubic-bezier).



Basic Calculator



Basic Operations

Add, subtract, multiply, divide with chained operation support.



Utilities

Sign toggle (+/-), percentage, decimal point, backspace delete.



4×4 Button Grid

Color-coded buttons: numbers, operators (blue), clear (red), delete (orange), equals (teal gradient).

Scientific Calculator



Trig & Log

$\sin$, $\cos$, $\tan$ (DEG/RAD toggle), $\log$, $\ln$

Math Functions

$\sqrt{}$, x^2 , x^y , $1/x$, $|x|$

Constants & Paren

π , e , and parentheses toggle for expression grouping.

Shared CalcEngine

A single reusable `CalcEngine(displayId, exprId)` class powers both calculators. Instances: `mc` and `sc`.

01

State

Tracks `current`, `stored`, `operator`, `freshNum`, and expression string.

02

Core Methods

`digit()`, `dot()`, `op()`, `equals()`, `clear()`,
`del()`, `toggleSign()`, `percent()`

03

Scientific Extensions

`fn()` handles all math functions;
`const()` injects π/e ; `toggleDeg()`
switches angle mode.

Display & Error Handling

Dual-Line Display

Top line shows expression (`.display-expr`), main line shows current value (`.display-main`) in Orbitron with cyan glow.

Number Formatting

`numStr()` uses `toPrecision(10)`, strips trailing zeros, handles $\pm\infty$ and NaN.

Error States

Div-by-zero, domain errors, undefined tan → red display, `.error` class, resets state.



Button Styling



Numbers

Translucent white, subtle hover lift, `:active` scale-down.



Operators

Blue-tinted (`rgba(0,150,255,0.15)`), larger font, glow on hover.



Equals

Gradient teal-to-cyan background, bold text, text-shadow glow.



Scientific

Purple-tinted, smaller font (0.78rem), letter-spacing for compact labels.

Keyboard Support & Summary

Keyboard Map

Active calculator listens to `keydown`: digits 0–9, `.`, `+` `-` `*` `/`, `Enter/=`, `Backspace`, `Escape`, `%`.

 Only the visible calculator responds – detected via `style.display === 'block'`.

Key Takeaways



Single `CalcEngine` class shared across both modes



Glass-morphism + particle system for premium feel



Full DEG/RAD trig, domain-error handling, 10-digit precision

Demo Calculator for Use

Calculator Link:

Click Here → <https://nikhilroy1356.github.io/Calculator-Project/>

Any Questions?

Thank You